Understanding of Non-Static:

Non-Static:

Any Member declared in the class block without prefixed with Static is known as Non-static Member of the class.

NOTE:

Non-Static Members get their memory allocation in the instance of class(object of class).

Therefore we can use the Non-Static Members from static context only with the help of address of an object.

Instance of class(object):

Object is the block of memory created during Runtime of program in the heap area.

Syntax:

new Costructor_of_the_class

Consructor Syntax:

class_Name();

new:

new is keyword.

new is unary operator which is used to create an object for a specific class/

new operator returns the address of the object created.

NOTE:

1. We can create any number of objects for the class.
2. Every time a new operator is used, a new object is created for the class.

Non-Static Variables:

A Global Variable not prefixed with the static modifier is known as Non-Static Variables.

NOTE:

Non-Static Variable will be allocated inside the every object of class which is created.

NOTE:

A Non-Static Member cannot be used directly or with help of class name inside the static context, if we try we get compile time error.

How to use a Non-Static Members with the static context:

-> with help of object reference.

Reference Variable:

The Variable which is used to call Non-Static Members.

It returns the address of an Object.

Syntax:

```
className identifier= new className();
```

How to use a Non-Static Members with the static context:

-> with help of object reference.

Reference Variable:

The Variable which is used to call Non-Static Members.

It returns the address of an Object.

Syntax:

```
className identifier= new className();
```

Multiple Objects for a class:

```
className Object1=new className();
className Object2=new className();
```

Note:

Non-static Variable is not a shared memory, every object will have it's own independent memory allocation.

Non-Static Method:

A Method without prefix with Static Modifier is known as Non-static Method.

Note:

The reference of the Non-Static methods will be stored inside the Objects.

The block which is belongs to the Non-Static method is known as Non-Static Context.

Inside the Non-Static context we can use the Non-Static Members of the class directly.

A Non-Static methods can be called from the Static context with help of object reference only.

We can call Non-Static Method inside the current class

-> With help of Object reference.

We can call Non-Static Method inside the different class

-> With help of Object reference.

Non-Static Context:

Any blocks belongs to Non-Static Member is Known as Non-Static Context.

Inside the Non-Static Context we can,

1.We Can use the Static Members of the same class directly.

2.We can use the Non-Static Members of the same class directly.

Inside the Static Context we can,

1.We Can use the Static Members of the same class directly.

2.We can use the Non-Static Members of the same class With help of Object reference.

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this:

It a keyword.

It is a Non-Static Member which will have the address of object which is currently used.

(address of current object).

Note:

We can use "this" only in Non-Static Context.

Purpose of using this(use of this):

It is used to access the member of the object from a Non-Static context.

If a Global and Local variable name is same we can use the object variable

with the help of this keyword.
